

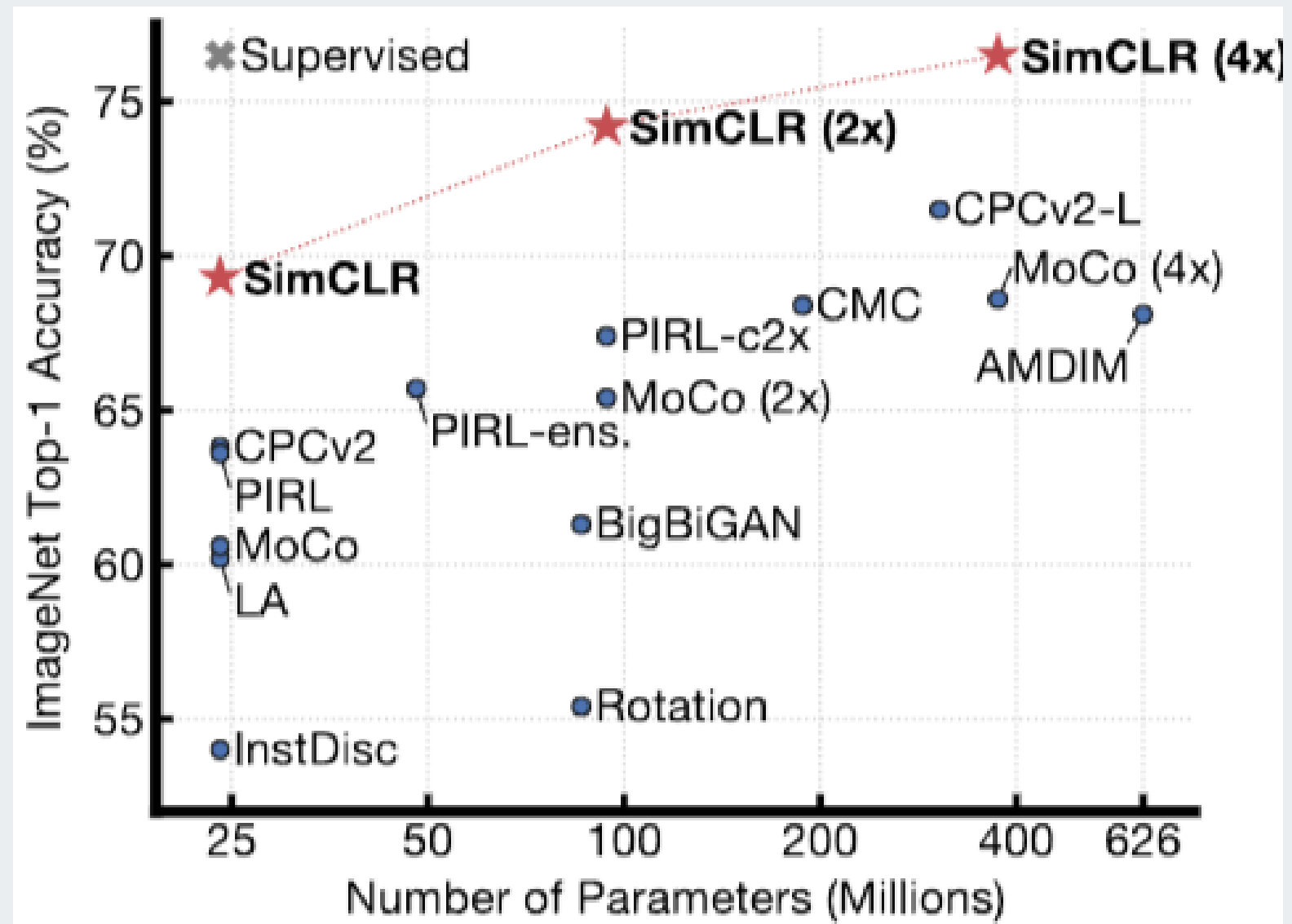
A **Simple** Framework for **Contrastive** Learning of Visual **Representations**

SimCLR

reference: <https://amitness.com/posts/simclr>

Presenter: Jiwon Kang

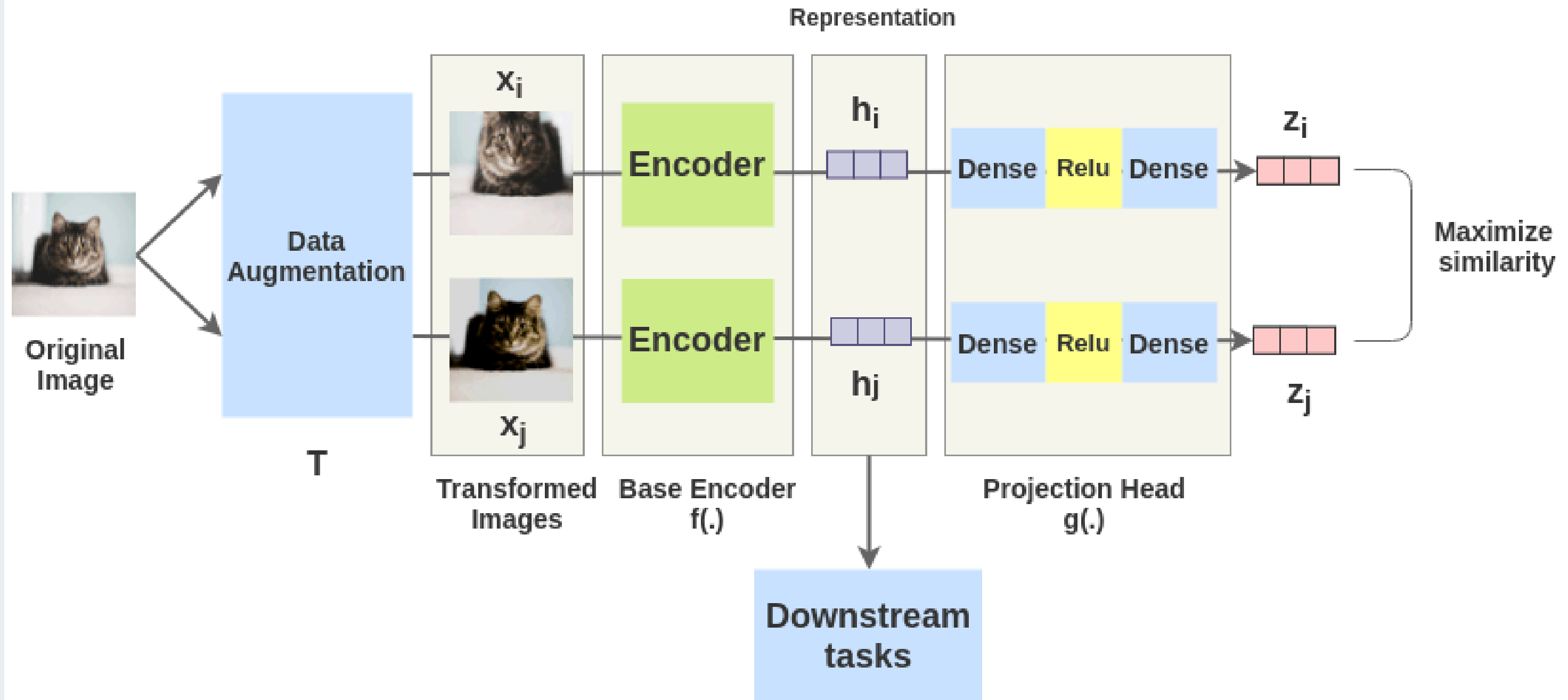
SimCLR



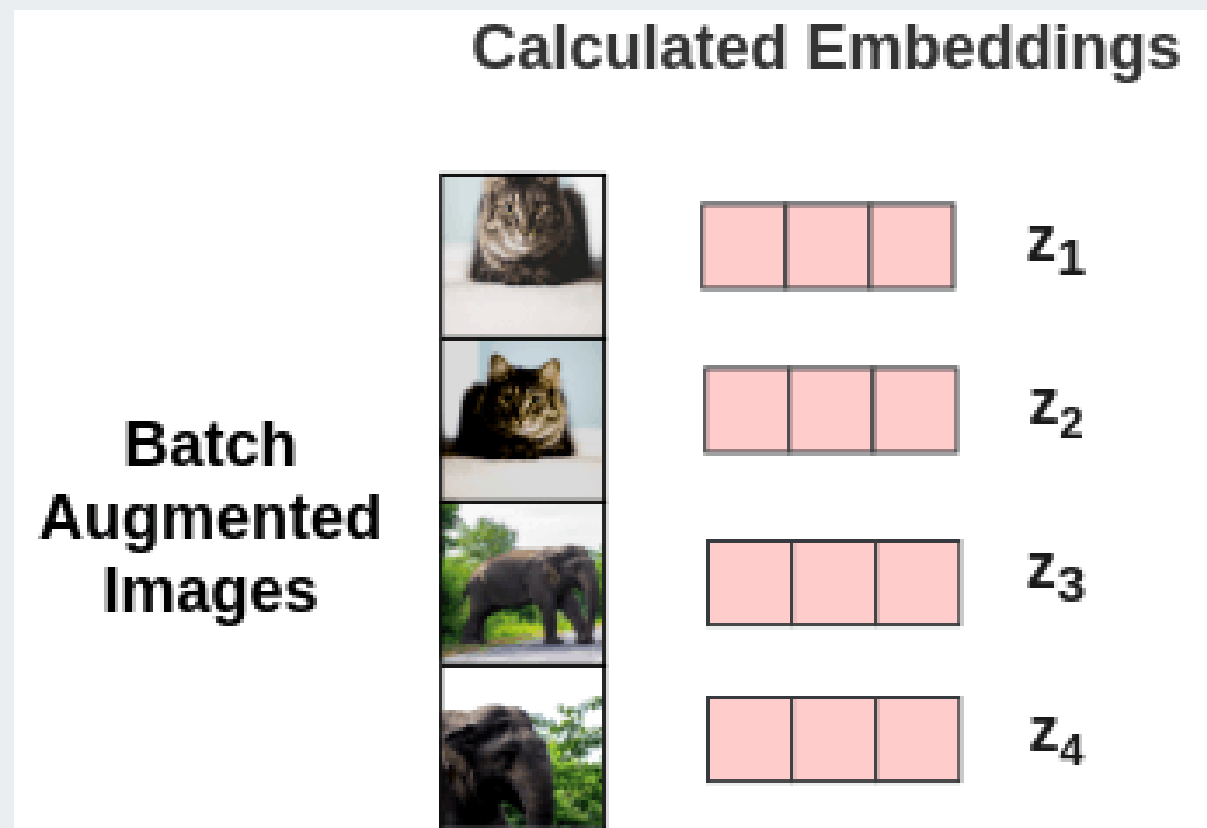
- Label 없이 representation 학습이 가능할까?
- 기존: Supervised Learning 의존

SimCLR

SimCLR Framework



SimCLR

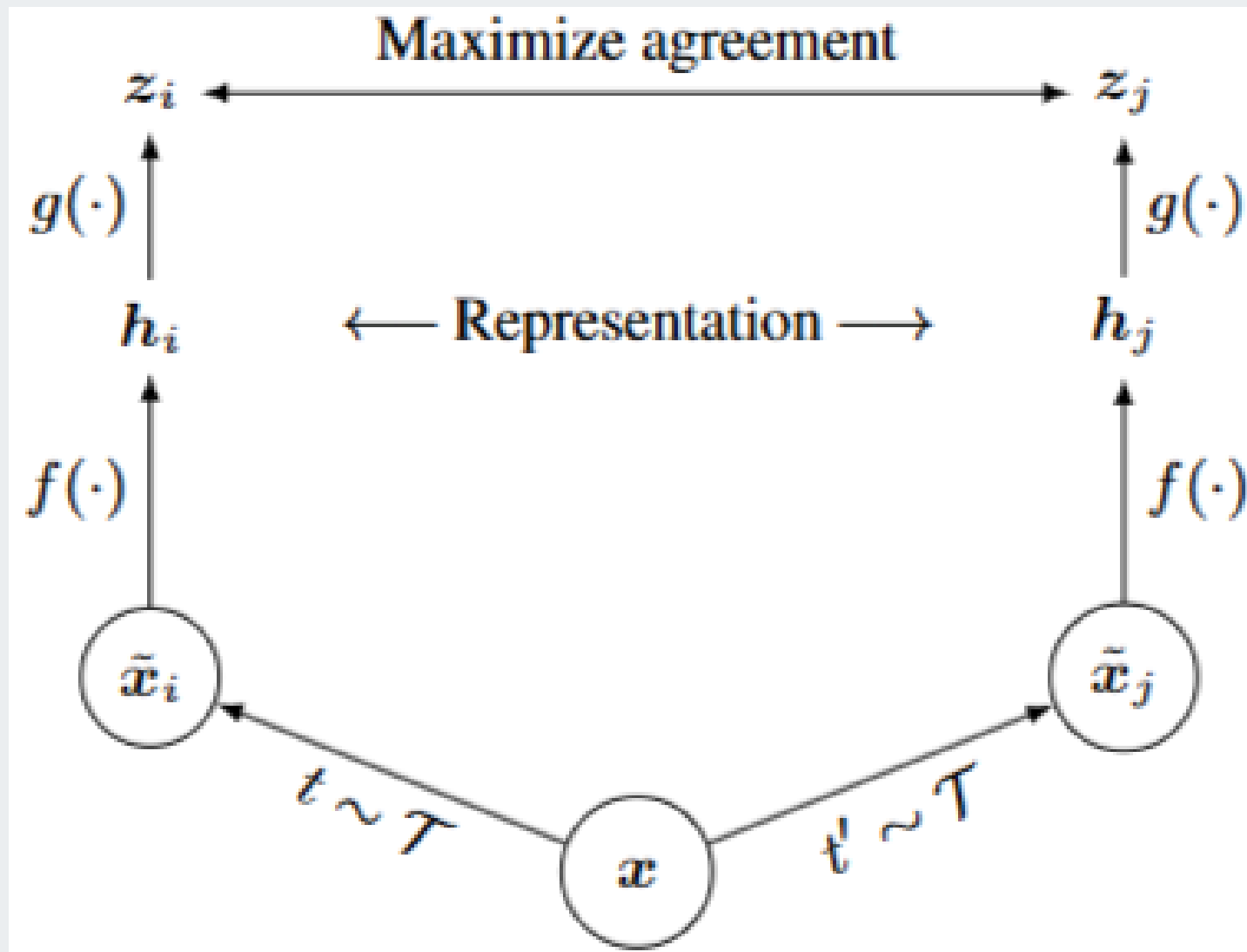


Similarity Calculation of Augmented Images

$$\text{similarity}(\overset{x_i}{\text{cat}}, \overset{x_j}{\text{cat}}) = \text{cosine similarity} \left(\overset{z_i}{\text{red box}}, \overset{z_j}{\text{red box}} \right)$$

$$s_{i,j} = \frac{z_i^T z_j}{(\tau ||z_i|| ||z_j||)}$$

SimCLR



Algorithm 1 SimCLR's main learning algorithm.

input: batch size N , constant τ , structure of f , g , $\mathcal{T}$.

for sampled minibatch $\{x_k\}_{k=1}^N$ **do**

for all $k \in \{1, \dots, N\}$ **do**

 draw two augmentation functions $t \sim \mathcal{T}, t' \sim \mathcal{T}$

 # the first augmentation

$\tilde{x}_{2k-1} = t(x_k)$

$h_{2k-1} = f(\tilde{x}_{2k-1})$

 # representation

$z_{2k-1} = g(h_{2k-1})$

 # projection

 # the second augmentation

$\tilde{x}_{2k} = t'(x_k)$

$h_{2k} = f(\tilde{x}_{2k})$

 # representation

$z_{2k} = g(h_{2k})$

 # projection

end for

for all $i \in \{1, \dots, 2N\}$ and $j \in \{1, \dots, 2N\}$ **do**

$s_{i,j} = z_i^\top z_j / (\|z_i\| \|z_j\|)$ # pairwise similarity

end for

define $\ell(i, j)$ **as** $\ell(i, j) = -\log \frac{\exp(s_{i,j}/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(s_{i,k}/\tau)}$

$\mathcal{L} = \frac{1}{2N} \sum_{k=1}^N [\ell(2k-1, 2k) + \ell(2k, 2k-1)]$

 update networks f and g to minimize $\mathcal{L}$

end for

return encoder network $f(\cdot)$, and throw away $g(\cdot)$

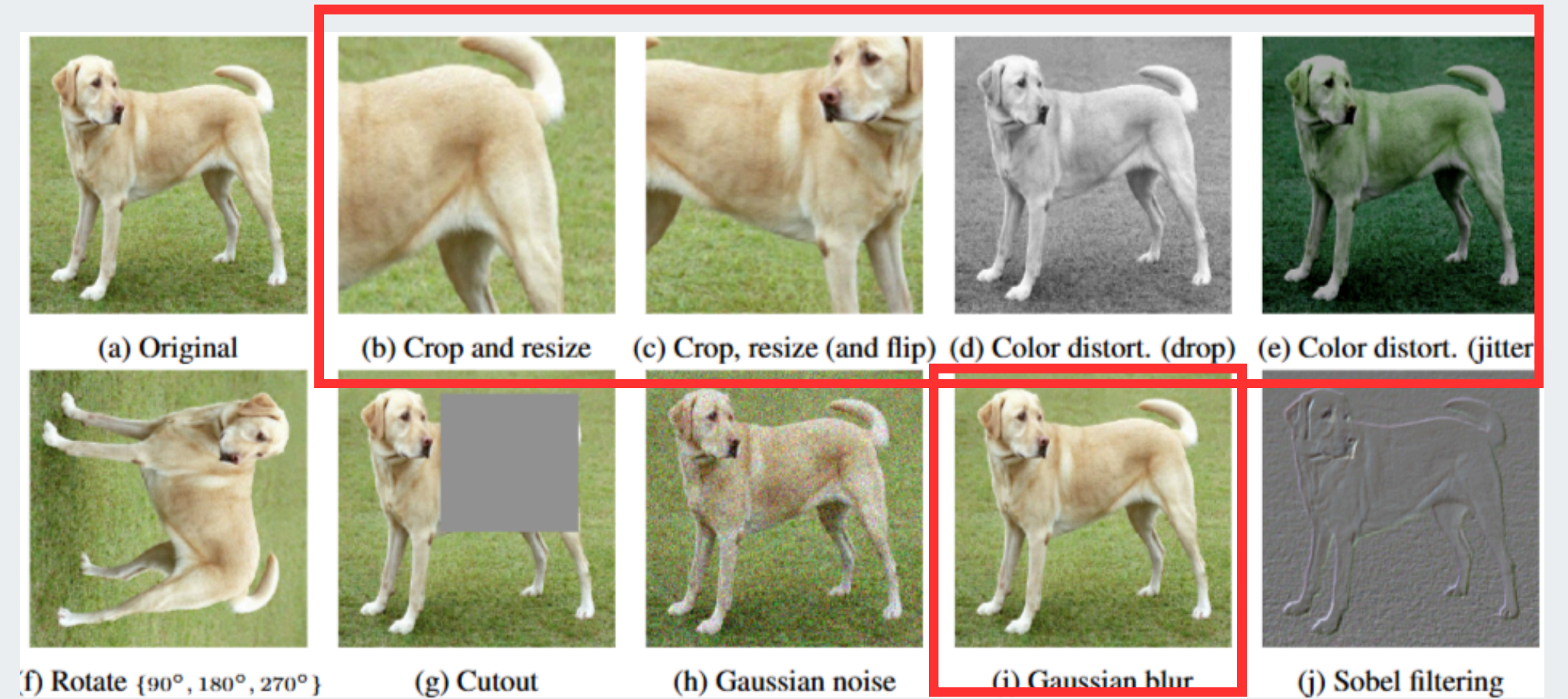
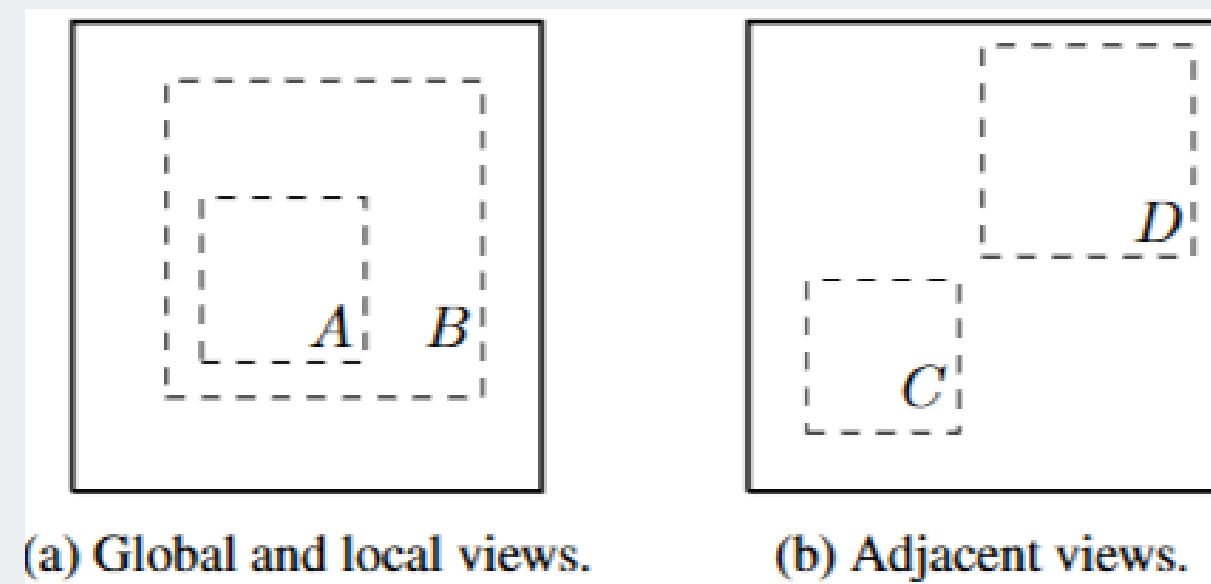
SimCLR

- Loss Function: **NT-Xent**
- Training Batch Size: 256~8192
- Default: 4096
- Dataset: ImageNet 2012
- Optimizer: **LARS** optimizer

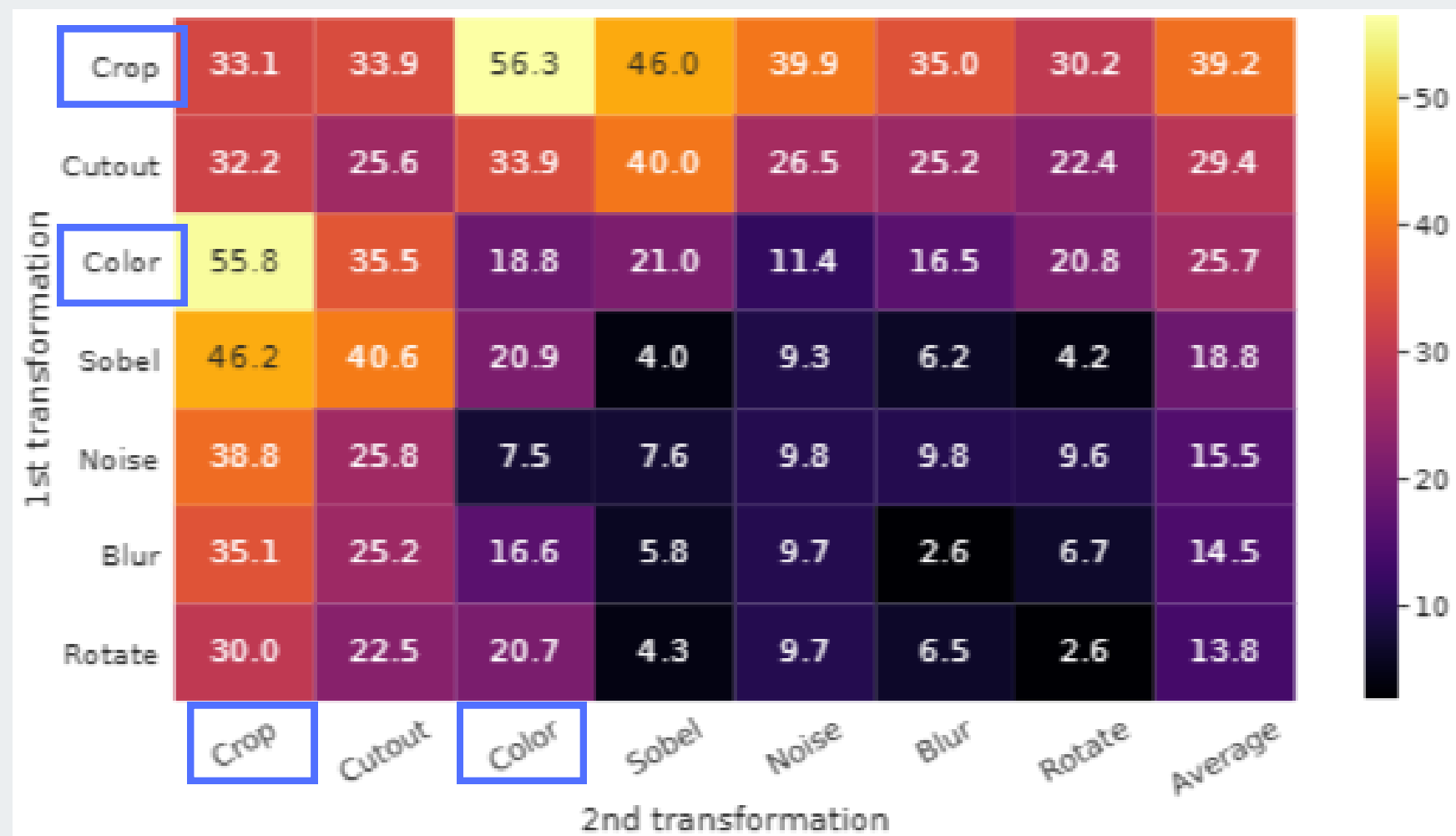
Algorithm 1 SimCLR's main learning algorithm.

```
input: batch size  $N$ , constant  $\tau$ , structure of  $f, g, \mathcal{T}$ .
for sampled minibatch  $\{x_k\}_{k=1}^N$  do
  for all  $k \in \{1, \dots, N\}$  do
    draw two augmentation functions  $t \sim \mathcal{T}, t' \sim \mathcal{T}$ 
    # the first augmentation
     $\tilde{x}_{2k-1} = t(x_k)$ 
     $h_{2k-1} = f(\tilde{x}_{2k-1})$  # representation
     $z_{2k-1} = g(h_{2k-1})$  # projection
    # the second augmentation
     $\tilde{x}_{2k} = t'(x_k)$ 
     $h_{2k} = f(\tilde{x}_{2k})$  # representation
     $z_{2k} = g(h_{2k})$  # projection
  end for
  for all  $i \in \{1, \dots, 2N\}$  and  $j \in \{1, \dots, 2N\}$  do
     $s_{i,j} = z_i^\top z_j / (\|z_i\| \|z_j\|)$  # pairwise similarity
  end for
  define  $\ell(i, j)$  as  $\ell(i, j) = -\log \frac{\exp(s_{i,j}/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(s_{i,k}/\tau)}$ 
   $\mathcal{L} = \frac{1}{2N} \sum_{k=1}^N [\ell(2k-1, 2k) + \ell(2k, 2k-1)]$ 
  update networks  $f$  and  $g$  to minimize  $\mathcal{L}$ 
end for
return encoder network  $f(\cdot)$ , and throw away  $g(\cdot)$ 
```

Data Augmentation



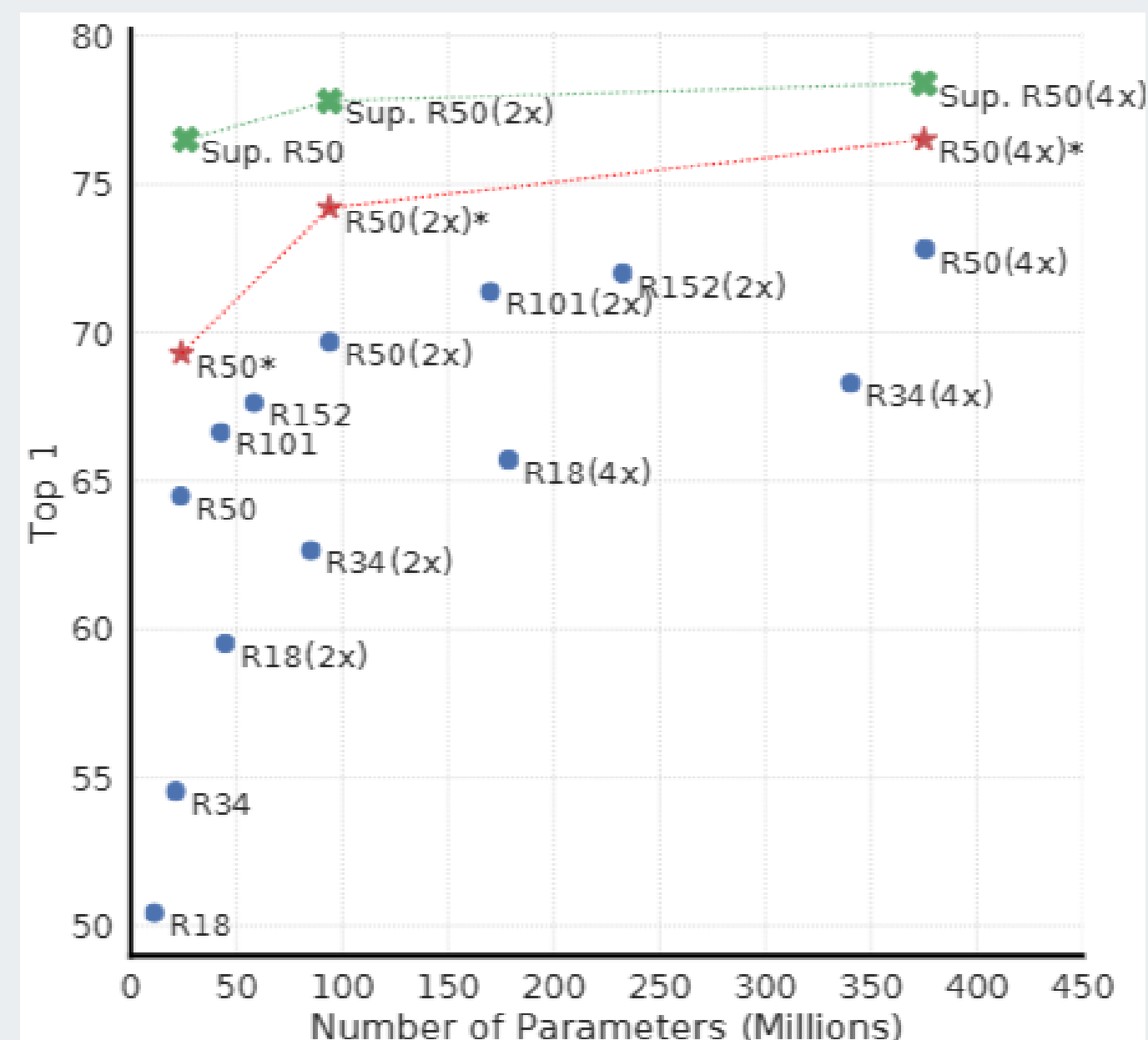
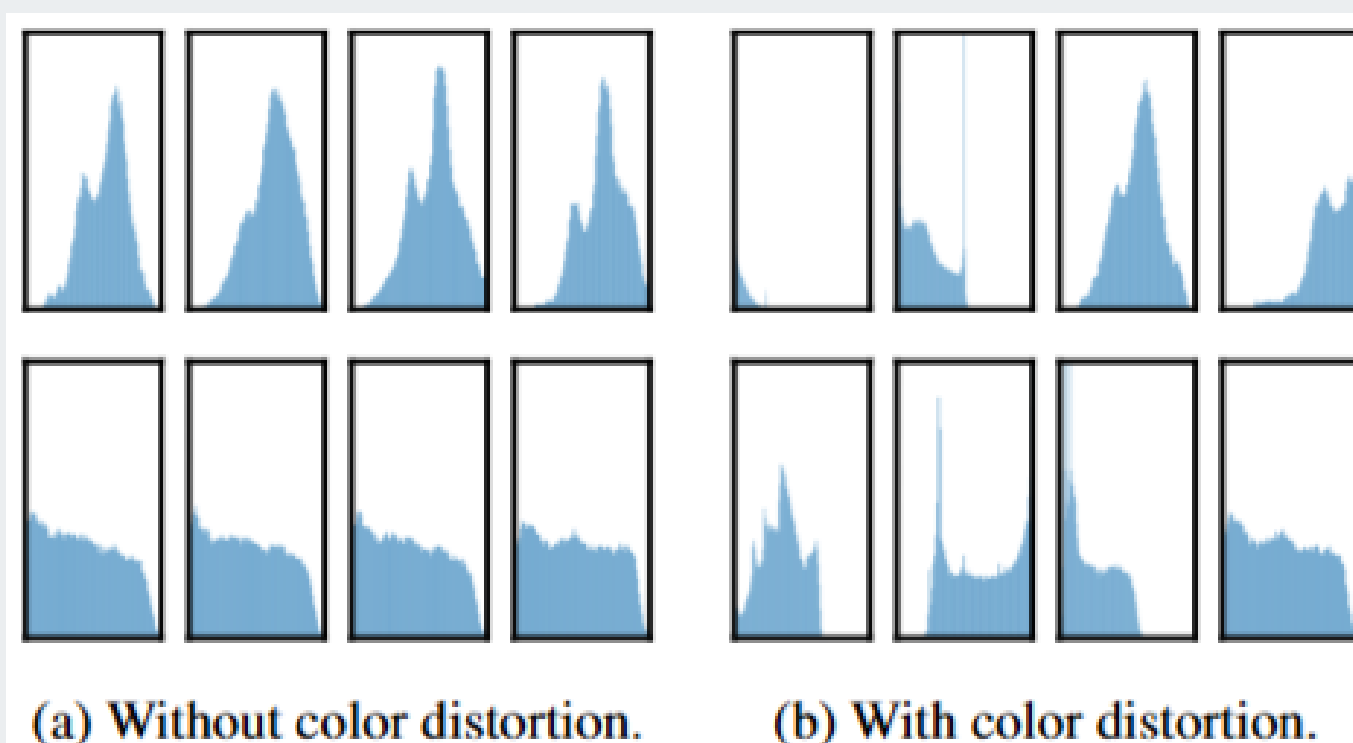
Data Augmentation



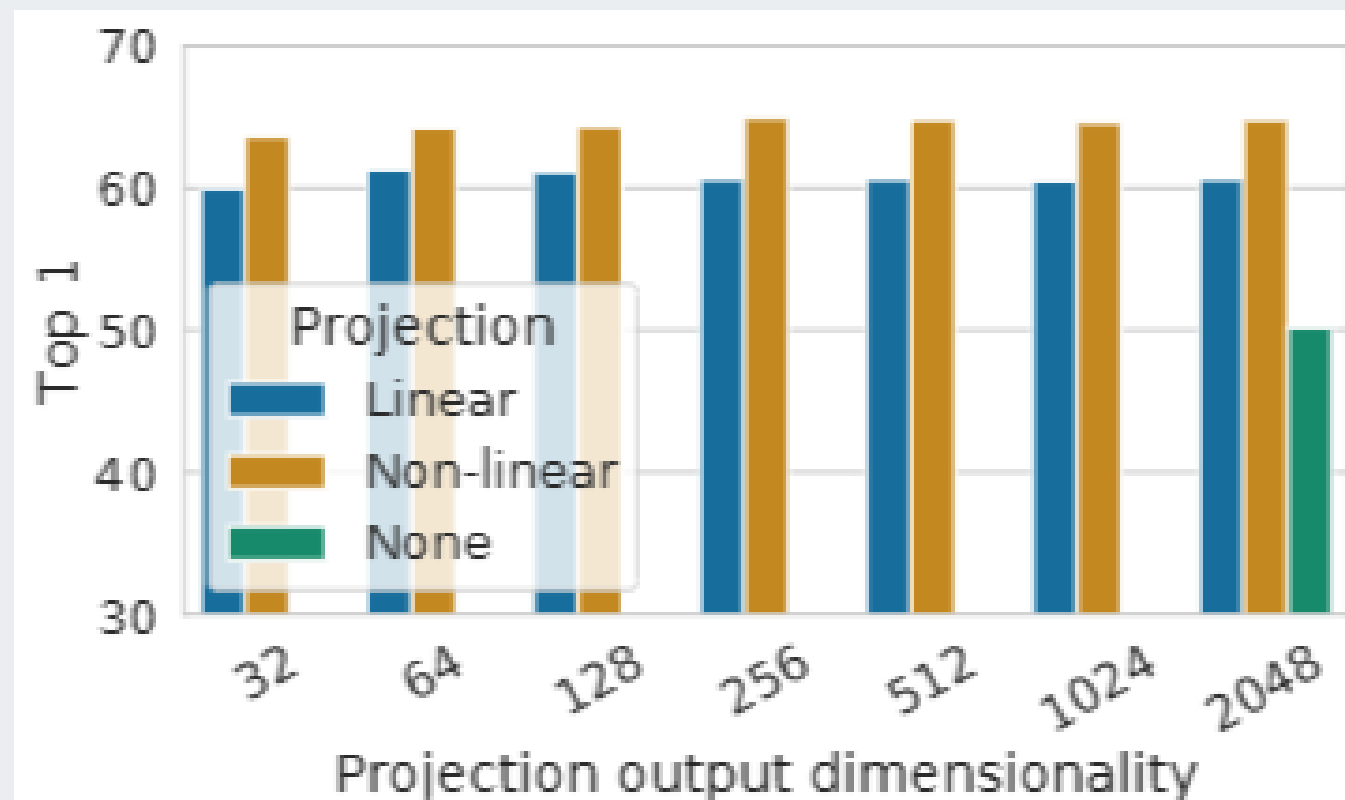
- 여러개의 augmentation 기법을 이어붙였을 때 더 좋은 성능 보임
- Crop, Color distort 조합이 가장 좋은 성능

Data Augmentation

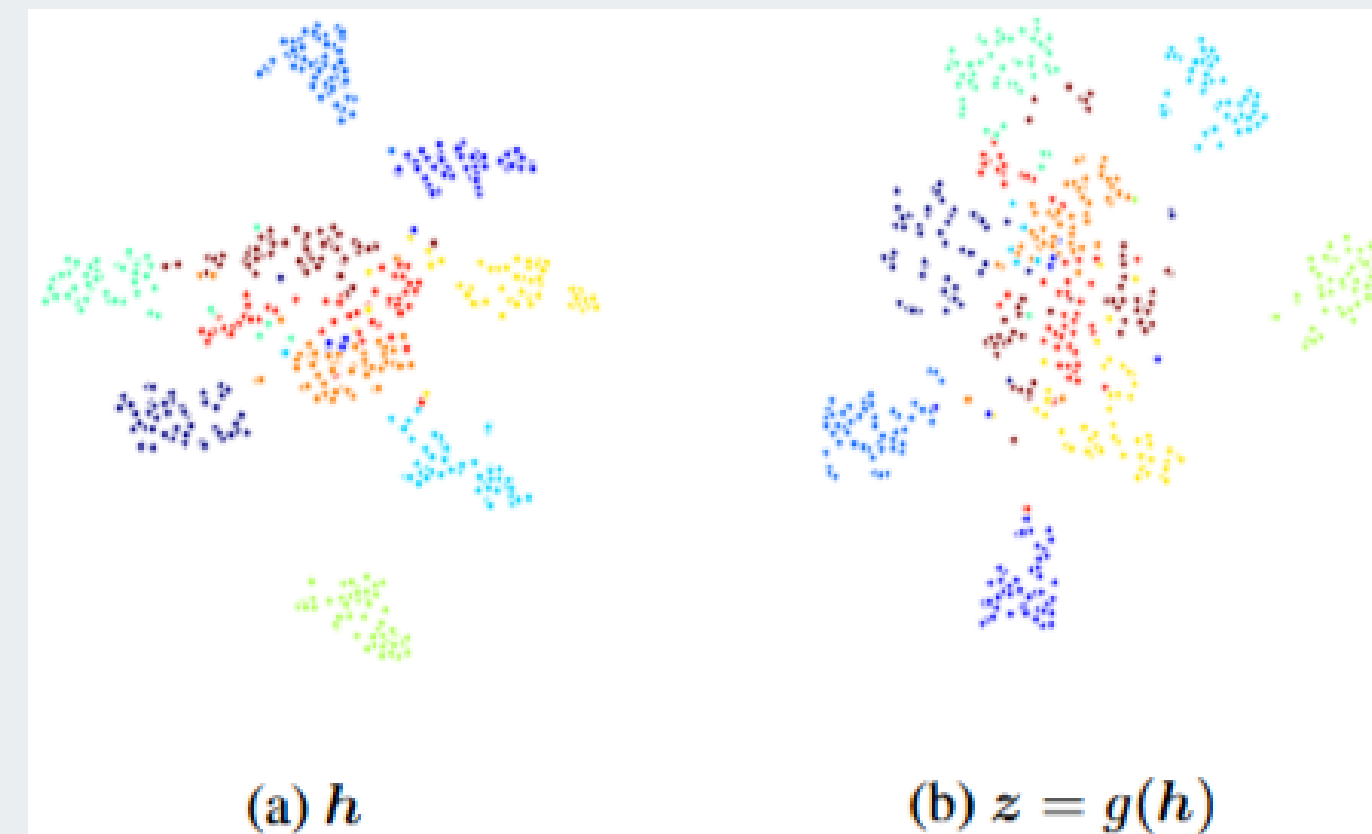
Methods	Color distortion strength					AutoAug
	1/8	1/4	1/2	1	1 (+Blur)	
SimCLR	59.6	61.0	62.6	63.2	64.5	61.1
Supervised	77.0	76.7	76.5	75.7	75.4	77.1



Non-linear Projection head



- Projection head(FC)를 사용하지 않았을때 보다 사용했을때 성능 더 좋음
- Non-linear에서 더 좋은 성능 보임



- encoder만 사용했을때, class별로 더 잘 구분됨

What to predict?	Random guess	Representation	
		h	$g(h)$
Color vs grayscale	80	99.3	97.4
Rotation	25	67.6	25.6
Orig. vs corrupted	50	99.5	59.6
Orig. vs Sobel filtered	50	96.6	56.3

Loss function

Name	Negative loss function	Gradient w.r.t. u
NT-Xent	$u^T v^+ / \tau - \log \sum_{v \in \{v^+, v^-\}} \exp(u^T v / \tau)$	$(1 - \frac{\exp(u^T v^+ / \tau)}{Z(u)}) / \tau v^+ - \sum_{v^-} \frac{\exp(u^T v^- / \tau)}{Z(u)} / \tau v^-$
NT-Logistic	$\log \sigma(u^T v^+ / \tau) + \log \sigma(-u^T v^- / \tau)$	$(\sigma(-u^T v^+ / \tau)) / \tau v^+ - \sigma(u^T v^- / \tau) / \tau v^-$
Margin Triplet	$-\max(u^T v^- - u^T v^+ + m, 0)$	$v^+ - v^-$ if $u^T v^+ - u^T v^- < m$ else 0

Margin	NT-Logi.	Margin (sh)	NT-Logi.(sh)	NT-Xent
50.9	51.6	57.5	57.9	63.9

	Food	CIFAR10	CIFAR100	Birdsnap	SUN397	Cars	Aircraft	VOC2007	DTD	Pets	Caltech-101	Flowers
<i>Linear evaluation:</i>												
SimCLR (ours)	76.9	95.3	80.2	48.4	65.9	60.0	61.2	84.2	78.9	89.2	93.9	95.0
Supervised	75.2	95.7	81.2	56.4	64.9	68.8	63.8	83.8	78.7	92.3	94.1	94.2
<i>Fine-tuned:</i>												
SimCLR (ours)	89.4	98.6	89.0	78.2	68.1	92.1	87.0	86.6	77.8	92.1	94.1	97.6
Supervised	88.7	98.3	88.7	77.8	67.0	91.4	88.0	86.5	78.8	93.2	94.2	98.0
Random init	88.3	96.0	81.9	77.0	53.7	91.3	84.8	69.4	64.1	82.7	72.5	92.5

Thank You for Listening